

1. A network shows an open circuit voltage of 9V and a short-circuit current of 1.5A at its terminals. Find R_{th} , I_N and the terminal voltage for a 4Ω load.

Sol: Open circuit voltage, $V_{th} = 9V$

Short circuit current, $I_N = I_{sc} = 1.5A$

Load Resistance, $R_L = 4\Omega$

To Find R_{th} ,

From Thevenin & Norton Relationship, $R_{th} = \frac{V_{th}}{I_N} = \frac{9}{1.5} = 6\Omega$

$\therefore R_N = R_{th} = 6\Omega$

Terminal voltage for 4Ω load,

Using Thevenin equivalent circuit,

$$V_L = \frac{V_{th} R_L}{R_{th} + R_L} = 9 \left(\frac{4}{6+4} \right) = 3.6V$$

2. A 100mH inductor & a 47μF Capacitor are placed in series. Find the frequency at which their reactance are equal.

Sol: Given,

$$L = 100mH = 0.1H$$

$$C = 47\mu F = 47 \times 10^{-6} F$$

At equal reactance, $X_L = X_C \Rightarrow \omega L = \frac{1}{\omega C}$

$$\therefore \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{(0.1)(47 \times 10^{-6})}} = 461.9 \text{ rad/s}$$

$$\text{Now, } f_0 = \frac{\omega_0}{2\pi} = \frac{461.9}{2\pi} = 73.5 \text{ Hz}$$

- 3: A inductor voltage waveform is $+20\text{V}$ for 3ms & $-V_2$ for the remaining 2ms of a 5ms period. Find V_2 .

Sol: Total period = 5ms

At steady state operation, $\int_0^T v_L(t) dt = 0$

$\therefore$ Positive & negative volt-seconds must be equal.

Apply volt-second balance,

$$(20)(3 \times 10^{-3}) + (-V_2)(2 \times 10^{-3}) = 0$$

$$\Rightarrow 60 \times 10^{-3} - 2V_2 \times 10^{-3} = 0$$

$$\Rightarrow V_2 = 30\text{V}$$

$\therefore$ Waveform of Inductor $+20\text{V}$ for 3ms & -30V for 2ms .

- 4: Sketch the impedance locus of a series RLC circuit as ω is swept from 0 to ∞ & identify the sketch at which the locus crosses the real axis.

Sol: For series RLC circuit,

$$Z = R + jX_L - jX_C$$

$$\therefore X_L = \omega L \text{ \& } X_C = \frac{1}{\omega C}$$

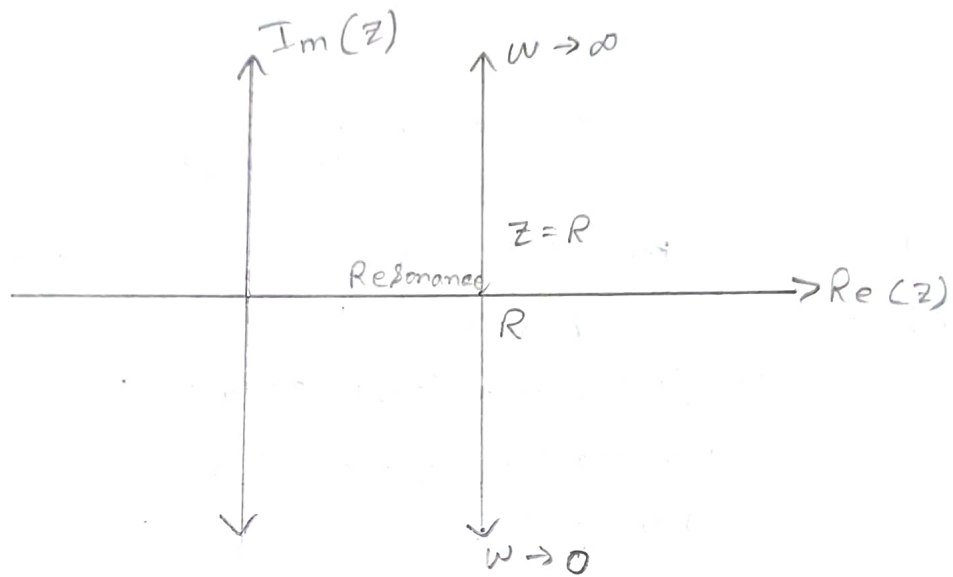
$$\therefore Z = R + j(\omega L - 1/\omega C)$$

$$\therefore \text{Re}(Z) = R \text{ \& } \text{Im}(Z) = \omega L - 1/\omega C$$

At $\omega = 0$, $X_L = 0$ but $X_C \rightarrow \infty$,

④ Impedance Locus

The locus is a vertical straight line at $\text{Re}(z) = R$:



$$\operatorname{Im}(Z) \rightarrow -\infty$$

So, the locus starts at $R - j\infty$

At resonance,

The locus crosses the real axis when $X_L = X_C$

$$\omega_0 L = 1/\omega_0 C \Rightarrow \omega_0 = 1/\sqrt{LC}$$

$$X_L - X_C = 0 \Rightarrow Z = R$$

At $\omega \rightarrow \infty$, $X_L \rightarrow \infty$ & $X_C \rightarrow 0$

$$\therefore \operatorname{Im}(Z) \rightarrow +\infty$$

So the locus goes toward $R + j\infty$.

- 5) Using Euler's Formula, express $v(t) = 20 \cos(314t + 30^\circ)$ as a phasor & state the frequency in Hz.

$$\text{Sol: } v(t) = 20 \cos(314t + 30^\circ)$$

Standard Sinusoidal form, $v(t) = v_m \cos(\omega t + \phi)$

$$v_m = 20, \omega = 314 \frac{\text{rad/sec}}, \phi = 30^\circ$$

$\therefore$ Phasor is $20 \angle 30^\circ \text{ V} = v'$

$$v_{\text{rms}} = \frac{20}{\sqrt{2}} = 14.14 \text{ V} \quad \text{Then, } v' = 14.14 \angle 30^\circ \text{ V}_{\text{rms}}$$

We know,

$$\omega = 2\pi f$$

$$\Rightarrow f = 314/2\pi$$

$$\Rightarrow f = 50 \text{ Hz}$$